2.3 Classwork: The Sine Rule and the Area of a Triangle — Markscheme

1. [Maximum mark: 11]

(a) $\frac{AC}{\sin 65^\circ} = \frac{8}{\sin 40^\circ}$ **M1**

$AC = \frac{8 \sin 65^\circ}{\sin 40^\circ}$

$AC = 11.3 \quad (11.2797\ldots)$ **A1**

M1 for a correct sine rule equation pairing each side with its opposite angle.

(b) $C = 180^\circ - 40^\circ - 65^\circ = 75^\circ$ **A1**

Area $= \frac{1}{2}(8)(11.2797\ldots)\sin 75^\circ$ **M1**

Area $= 43.6 \quad (43.5814\ldots)$ **A1**

M1 for $\frac{1}{2}ab\sin C$ *with two sides and the angle between them. Accept* 43.7 *from the rounded* $AC = 11.3$*. FT from (a).*

(c) $\frac{\sin R}{10} = \frac{\sin 35^\circ}{8}$ **M1**

$\sin R = \frac{10 \sin 35^\circ}{8} = 0.716970\ldots$

$R = 45.8^\circ \quad (45.8049\ldots)$ **A1**

The obtuse value 134.2° *is excluded by the statement that* R *is acute.*

(d) Area $= \frac{1}{2}(6)(9)\sin 30^\circ$ **M1**

Area $= \frac{27}{2}$ **A1**

Accept 13.5*.*

(e) $\frac{1}{2}(5)(10)\sin\theta = 20$ **M1**

$\sin\theta = 0.8$

$\theta = 53.1^\circ \quad (53.1301\ldots)$ **A1**

2. [Maximum mark: 9]

(a) $\angle ATB = 180° - 52° - 71° = 57°$ **A1**

(b) $\frac{AT}{\sin 71°} = \frac{200}{\sin 57°}$ **M1**

$AT = 225\,\text{m}$ $(225.480\ldots)$ **A1**

FT from (a).

(c) width $= AT \sin 52°$ **M1**

width $= 178\,\text{m}$ $(177.680\ldots)$ **A1**

M1 for right-angled trigonometry in the triangle formed by T, A and the foot of the perpendicular from T. Accept $BT \sin 71°$ with $BT = 187.92\ldots$ from the sine rule. Accept 177 from the rounded $AT = 225$. FT from (b).

(d) Area $= \frac{1}{2}(200)(225.480\ldots)\sin 52°$ **M1**

Area $= 17\,800\,\text{m}^2$ $(17\,768.0\ldots)$ **A1**

Accept $\frac{1}{2}(200)$(width). Accept 17 700 from the rounded $AT = 225$. FT from (b) or (c).

(e) $\frac{1}{2}(8)(10)\sin\theta = 30$ **M1**

$\sin\theta = 0.75$

$\theta = 48.6°$ $(48.5903\ldots)$ **A1**